

BODO AGDM, AB, Canada

WMO: 712140

Lat: 52.1242N

Lon: 110.1014W

Elev: 2254

StdP: 13.54

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.7	-16.5	-28.2	1.2	-21.2	-22.6	1.7	-16.2	26.0	22.7	22.9	20.6	7.1	300	0.702

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	86.6	63.2	82.7	62.4	79.2	61.0	66.8	79.7	64.8	77.5	62.9	75.1	11.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	91.4	71.9	60.3	85.0	69.4	58.4	79.1	67.4	32.7	79.8	31.0	77.3	29.6	74.6	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	21.2	18.9	DB	-28.2	92.8	4.5	3.6	-31.4	95.4	-34.1	97.5	-36.6	99.5	-39.9	102.1	(4)
(5)			WB	-28.1	70.6	4.6	2.5	-31.4	72.4	-34.1	73.9	-36.7	75.3	-40.0	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	13.1	12.4	25.1	39.9	51.7	59.2	64.6	62.4	53.5	40.3	24.6	12.7
	DBStd	21.99	14.87	13.57	13.50	9.69	7.92	6.01	5.59	6.52	8.58	9.62	12.20	14.25
	HDD50	5672	1144	1052	772	321	75	5	0	2	59	323	764	1155
	HDD65	9839	1609	1472	1237	754	416	193	76	128	356	765	1213	1620
	CDD50	1453	0	0	1	17	128	280	453	387	163	23	1	0
	CDD65	142	0	0	0	0	3	18	63	48	10	0	0	0
	CDH74	2243	0	0	1	5	115	266	861	715	269	11	0	0
	CDH80	646	0	0	0	0	16	60	264	215	91	0	0	0
Wind	WSAvg	9.3	9.2	8.8	9.9	10.7	10.1	9.3	8.6	8.4	8.8	9.7	9.4	8.7
Precipitation	PrecAvg	15.2	0.7	0.4	0.6	1.1	1.6	2.8	2.8	2.0	1.1	0.7	0.6	0.6
	PrecMax	20.6	1.5	0.9	1.4	3.2	3.4	4.7	5.4	5.0	2.9	1.6	2.0	1.1
	PrecMin	9.5	0.2	0.2	0.0	0.1	0.2	0.9	1.0	0.2	0.0	0.1	0.1	0.2
	PrecStd	3.0	0.4	0.2	0.3	0.7	0.9	1.1	1.0	1.0	0.9	0.4	0.5	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.5	43.2	59.2	73.2	82.5	86.5	91.0	90.9	88.9	75.2	57.1	45.1
		MCWB	37.8	37.2	46.1	51.3	57.0	62.2	65.4	63.5	60.3	55.2	44.0	38.1
	2%	DB	40.2	39.1	52.0	67.6	77.3	80.7	86.6	85.7	81.7	66.6	49.6	40.3
		MCWB	35.2	34.4	42.2	49.0	55.1	60.8	65.4	62.6	58.6	51.1	40.8	35.0
	5%	DB	37.1	35.1	46.3	61.7	72.9	76.5	82.5	81.6	75.4	60.7	44.1	36.5
		MCWB	33.1	31.6	38.9	45.8	53.6	59.4	64.3	62.2	56.5	47.9	37.4	32.2
	10%	DB	33.5	31.7	41.7	56.7	68.4	72.9	78.7	77.8	69.7	55.3	39.7	32.0
		MCWB	30.5	29.1	36.3	43.5	51.5	57.6	63.1	60.9	54.2	44.7	34.8	29.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	37.8	46.9	52.7	59.4	66.4	71.0	68.3	62.4	55.9	45.1	39.2
		MCDB	42.8	43.1	58.0	71.2	76.7	79.2	83.8	80.7	80.7	72.7	54.7	44.3
	2%	WB	35.6	34.5	42.2	49.5	56.6	63.1	67.9	65.8	59.9	52.0	41.5	35.3
		MCDB	39.8	38.3	50.4	65.4	73.3	75.9	80.9	79.3	77.0	64.2	48.7	39.8
	5%	WB	33.3	32.0	39.2	46.7	54.7	61.1	65.9	63.8	57.6	48.8	38.0	32.3
		MCDB	36.9	34.8	45.4	59.9	69.9	73.3	78.3	77.0	72.5	58.6	42.9	36.1
	10%	WB	30.5	28.8	36.4	44.1	52.6	59.1	64.0	61.8	55.4	45.8	35.2	29.1
		MCDB	33.4	31.5	41.6	54.8	66.4	70.1	75.9	74.2	68.9	54.8	39.3	32.0
Mean Daily Temperature Range	5% DB	MDBR	17.7	17.9	17.9	21.5	23.6	21.4	23.6	24.0	23.7	21.0	16.6	16.4
		MCDBR	19.1	20.5	23.5	30.4	30.0	27.5	29.8	30.9	33.9	29.6	22.0	20.3
		MCWBR	15.0	16.5	14.9	15.3	13.6	12.3	12.6	12.7	14.6	15.9	14.6	15.7
	5% WB	MCDBR	19.5	20.2	22.1	28.5	26.8	24.3	26.1	26.9	30.7	28.3	20.9	19.7
		MCWBR	15.8	16.6	14.7	15.1	12.9	12.3	12.6	12.4	14.1	15.9	14.4	15.6
Clear-Sky Solar Irradiance	taub		0.253	0.284	0.293	0.341	0.369	0.374	0.368	0.381	0.325	0.287	0.261	0.241
	taud		2.338	2.292	2.423	2.326	2.294	2.330	2.381	2.313	2.441	2.526	2.457	2.366
	Ebn at Noon		245	266	289	284	280	279	279	267	273	266	243	231
	Edh at Noon		19	27	29	36	39	38	36	36	28	21	17	15
All-Sky Solar Radiation	RadAvg		357	692	1143	1518	1847	1905	2025	1643	1173	702	361	265
	RadStd		34	54	89	138	129	134	98	132	116	95	37	26

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page